



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
TRIAL EXAMINATIONS 2006

FORM VI

MATHEMATICS EXTENSION 1

Examination date

Tuesday 8th August 2006

Time allowed

2 hours (plus 5 minutes reading time)

Instructions

- All seven questions may be attempted.
- All seven questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the seven questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

Checklist

- SGS booklets: 7 per boy. A total of 1000 booklets should be sufficient.
- Candidature: 120 boys.

Examiner

JNC

QUESTION ONE (12 marks) Use a separate writing booklet.

Marks

- (a) Find the exact value of $\tan^{-1}(-\sqrt{3})$. 1
- (b) Differentiate $e^{2x} \sin x$. 2
- (c) Find the exact value of $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx$. 2
- (d) Find the acute angle, correct to the nearest minute, between the lines $3x + y = 4$ and $x - y = 1$. 2
- (e) Given $A(2, 1)$ and $B(7, 3)$, find the coordinates of the point C which divides the interval AB externally in the ratio $2 : 3$. 2
- (f) Use the substitution $u = x + 1$ to find $\int x(x + 1)^3 \, dx$. 3

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

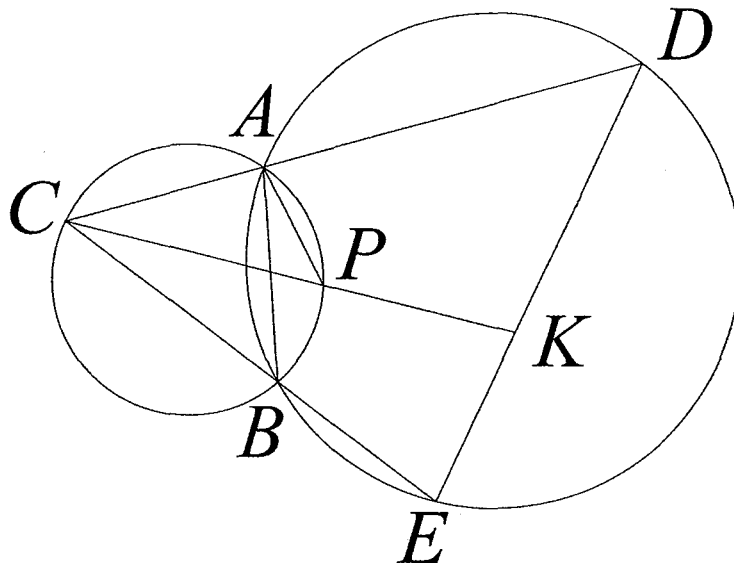
- (a) Solve $\frac{3}{x-1} \leq 2$. 3
- (b) Find the value of h if $x - 2$ is a factor of $P(x) = 3x^2 - 2hx + 7$. 2
- (c) Consider the function $f(x) = 3 \cos^{-1} \frac{x}{2}$.
- (i) Evaluate $f(0)$. 1
- (ii) State the domain and range of $y = f(x)$. 2
- (iii) Sketch the graph of $f(x)$. 1
- (d) A particle executes simple harmonic motion about the origin with period T seconds and amplitude A centimetres. Find its maximum speed in terms of T and A . 3

QUESTION THREE (12 marks) Use a separate writing booklet.

Marks

- (a) Show that the equation of the normal to the parabola $x = 2at$, $y = at^2$ at the point where $t = T$ is given by $x + Ty = 2aT + aT^3$. 3

(b)



In the diagram above, the two circles intersect at A and B , and CAD , CBE , CPK and DKE are straight lines.

- (i) Give a reason why $\angle APC = \angle ABC$. 1
- (ii) Hence, or otherwise, show that $ADKP$ is a cyclic quadrilateral. 3
- (c) A cup of hot milk at temperature T° Celsius loses heat when placed in a cooler environment. It cools according to the law given by the differential equation

$$\frac{dT}{dt} = -k(T - S)$$

where t is the time elapsed in minutes, S is the temperature of the environment in degrees Celsius and k is a positive constant.

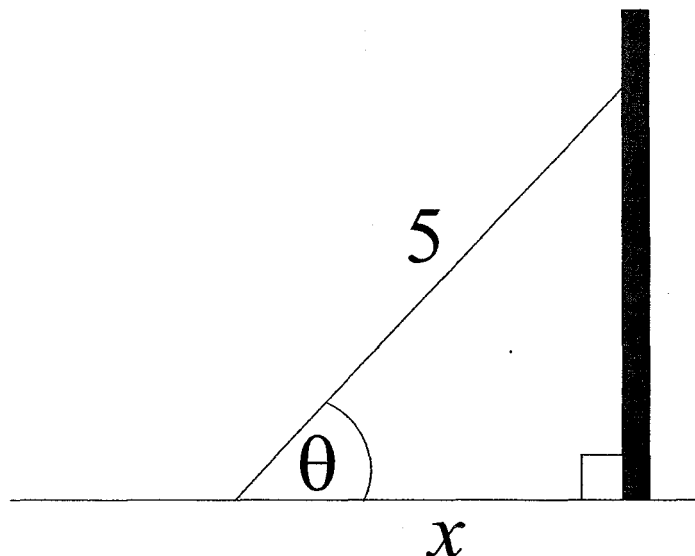
- (i) Show that $T = S + Ae^{-kt}$, where A is a constant, is a solution of the differential equation. 1
- (ii) (α) A cup of milk at 80°C is placed in an environment at 20°C , and after ten minutes it has cooled to 40°C . Find the exact value of k . 3
- (β) Find the temperature of the milk after five more minutes have elapsed. Give your answer rounded to the nearest tenth of a degree. 1

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

(a)

3



The diagram above shows a 5 metre ladder leaning against a wall on level ground. The base of the ladder is sliding away from the wall at 2 centimetres per second. Find the rate at which the angle of inclination θ is changing when the foot of the ladder is 3 metres from the wall.

(b) Find the coefficient of x^2 in the expansion of $\left(x^2 + \frac{2}{x}\right)^{10}$.

3

(c) (i) Taking about one-third of a page and on the same set of axes, draw sketches of $y = \ln x$ and $y = \sin x$ for $0 \leq x \leq 2\pi$.

2

(ii) On your diagram, indicate the root α of the equation $\ln x - \sin x = 0$.

1

(iii) Show that $\frac{\pi}{2} < \alpha < \frac{3\pi}{4}$.

1

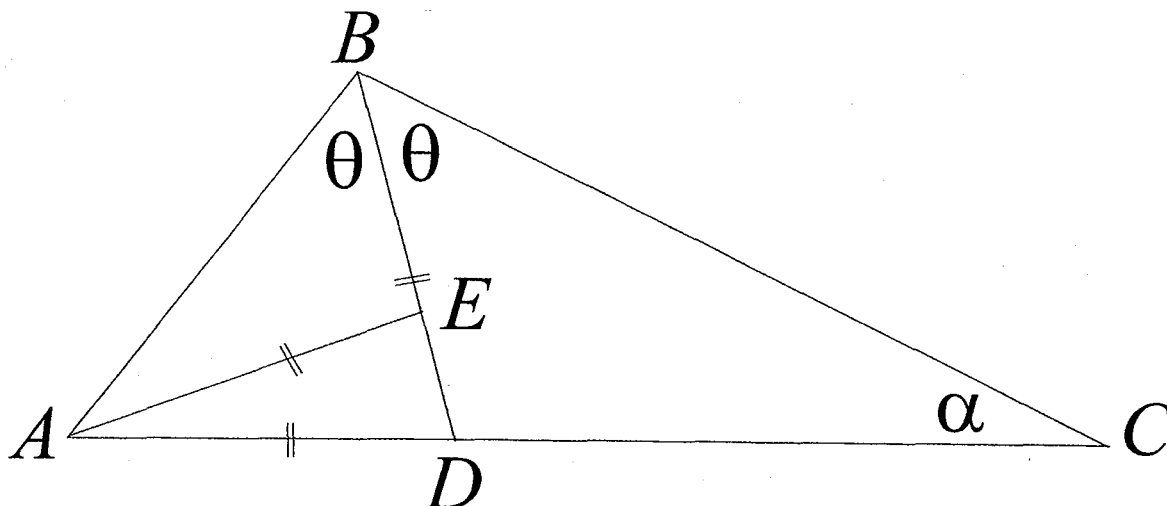
(iv) Use Newton's method once, with first approximation $x_1 = \frac{5\pi}{8}$, to find a better approximation for α . Give your answer correct to two decimal places.

2

QUESTION FIVE (12 marks) Use a separate writing booklet.

Marks

(a)



In the diagram above, ABD and AED are isosceles triangles with $AD = BD = AE$, and BD bisects $\angle ABC$. Let $\angle ABD = \angle CBD = \theta$ and let $\angle DCB = \alpha$.

- (i) Show that $\angle EAB = \alpha$, giving reasons. 3
 - (ii) Hence show that $\triangle ABE \parallel \triangle CBD$. 1
 - (iii) Deduce that $AE^2 = BE \times CD$. 2
- (b) (i) By squaring both sides, show that $2n + 3 > 2\sqrt{(n+1)(n+2)}$ for $n > 0$. 1
- (ii) Prove by mathematical induction that 5

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} > 2(\sqrt{n+1} - 1)$$

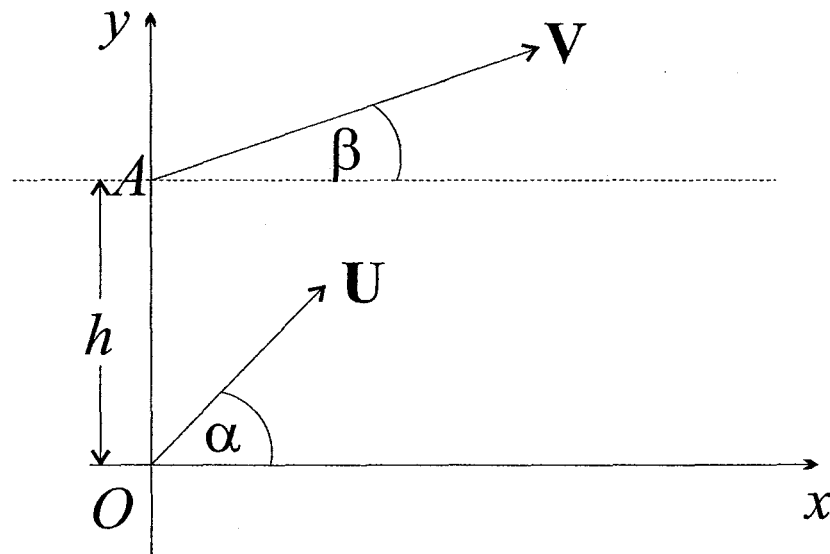
for all positive integer values of n .

QUESTION SIX (12 marks) Use a separate writing booklet.**Marks**

(a) By using the substitution $x = \tan \theta$, evaluate $\int_0^1 \frac{1}{(1+x^2)^{\frac{3}{2}}} dx$. 4

(b) When $(3+2x)^n$ is expanded as a polynomial in x , the coefficients of x^5 and x^6 are equal. Find the value of n . 4

(c) 4



In the diagram above, a particle is projected from the origin O with speed U metres per second at an angle of elevation α . At the same instant, another particle is projected from the point A , h metres directly above O , with speed V metres per second at an angle of elevation β , where $\beta < \alpha$. The particles move freely under gravity in the same plane of motion and collide T seconds after projection.

You may assume that the horizontal and vertical components of displacement at time t seconds of the particle projected from O are given by

$$x_O = Ut \cos \alpha \text{ and } y_O = Ut \sin \alpha - \frac{1}{2}gt^2 \text{ respectively.}$$

You may also assume that the horizontal and vertical components of displacement at time t seconds of the particle projected from A are given by

$$x_A = Vt \cos \beta \text{ and } y_A = h + Vt \sin \beta - \frac{1}{2}gt^2 \text{ respectively.}$$

Show that

$$T = \frac{h \cos \beta}{U \sin(\alpha - \beta)}.$$

QUESTION SEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) (i) Use the binomial theorem to find a simplified expansion for

1

$$(1+x)^{10n} + (1-x)^{10n},$$

where n is a positive integer.

- (ii) Hence evaluate

2

$$1 + \binom{30}{2} + \binom{30}{4} + \cdots + \binom{30}{30}.$$

- (b) Consider the function
- $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- .

- (i) Show that
- $f(x)$
- is an odd function.

1

- (ii) Examine the behaviour of
- $f(x)$
- as
- $x \rightarrow \infty$
- and as
- $x \rightarrow -\infty$
- .

2

- (iii) Show that the curve is increasing for all values of
- x
- .

2

- (iv) Sketch the curve
- $y = f(x)$
- .

1

- (v) If
- k
- is a positive constant, show that the area bounded by the curve
- $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- and the lines
- $x = 0$
- ,
- $x = k$
- and
- $y = 1$
- is always less than
- $\ln 2$
- .

3**END OF EXAMINATION**

FORM VI - EXTENSION I TRIAL - 2006

Question ONE

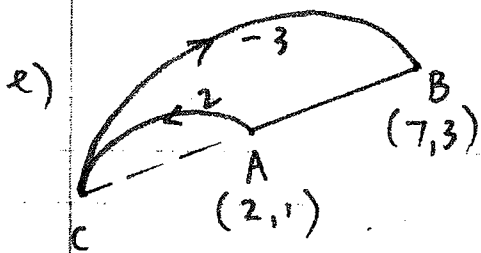
$$b) \frac{d}{dx} (e^{2x} \sin x) = 2e^{2x} \sin x + e^{2x} \cos x$$

a) $\tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$

$$\begin{aligned} \text{c) } \int_0^{\frac{\pi}{2}} \cos^2 x \, dx &= \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 2x) \, dx \\ &= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[\left(\frac{\pi}{2} + 0 \right) - 0 \right] \\ &= \frac{\pi}{4} \end{aligned}$$

$$d) \tan \theta = \frac{-3-1}{1+(-3)(+1)}$$

$$\therefore \theta = 63^\circ 26'$$



$$x = \frac{-6 + 14}{-1}$$
$$= -8 \checkmark$$

$$y = \frac{-3 + 6}{-1}$$
$$= -3$$

$$\therefore C = (-8, -3)$$

f) $\mu = x + 1$

$$\frac{du}{dx} = 1$$

$$\int x(x+1)^3 dx = \int (u-1)u^3 du$$

$$= \int u^4 - u^3 \, du$$

$$= \left[\frac{u^5}{5} - \frac{u^4}{4} \right] + C$$

$$= \frac{1}{5}(x+1)^5 - \frac{1}{4}(x+1)^4 + c$$

QUESTION TWO

a) $\frac{3}{x-1} \leq 2, x \neq 1$

$$3(x-1) \leq 2(x-1)^2 \quad \checkmark$$

$$(x-1)(2(x-1)-3) \geq 0$$

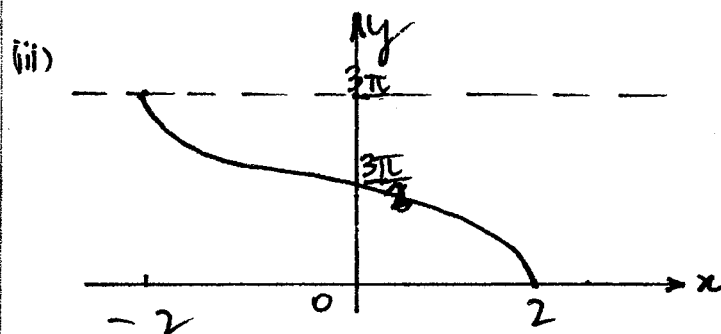
$$(x-1)(2x-5) \geq 0 \quad \checkmark$$

$$\therefore x < 1 \text{ or } x \geq \frac{5}{2} \quad \checkmark$$

b) $P(2) = 12 - 4h + 7 = 0 \quad \checkmark$
 $\therefore h = \frac{19}{4} \quad \checkmark$

c) (i) $f(0) = \frac{3\pi}{2} \quad \checkmark$

(ii) $-1 \leq \frac{x}{2} \leq 1 \quad \checkmark$ and $0 \leq \frac{y}{3} \leq \pi \quad \checkmark$
 $-2 \leq x \leq 2 \quad \checkmark$ $0 \leq y \leq 3\pi \quad \checkmark$



d) $\frac{2\pi}{n} = T$

$$\therefore n = \frac{2\pi}{T} \text{ and amplitude is } A$$

$$x = A \sin\left(\frac{2\pi}{T}t + \alpha\right) \quad \checkmark \quad \text{OR } v^2 = \left(\frac{2\pi}{T}\right)^2 (A^2 - x^2)$$

$$\dot{x} = \frac{2\pi A}{T} \cos\left(\frac{2\pi}{T}t + \alpha\right) \quad \checkmark \quad \text{Max at } x = 0;$$

Max velocity when

$$\cos\left(\frac{2\pi}{T}t + \alpha\right) = +1$$

$$\text{So max speed} = \frac{2\pi A}{T} \quad \checkmark$$

$$v^2 = \left(\frac{2\pi A}{T}\right)^2$$

$$v = \pm \frac{2\pi A}{T}$$

$$\text{So max speed is } \frac{2\pi A}{T}$$

QUESTION THREE

(a) $\left. \begin{aligned} x &= 2at \\ y &= at^2 \end{aligned} \right\}$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

$$= \frac{2at}{2a}$$

$$= t$$

when $t = T$, $\frac{dy}{dx} = T$, $x = 2aT$, $y = aT^2$ ✓

gradient of normal $= -\frac{1}{T}$ ✓

equation of normal: $y - aT^2 = -\frac{1}{T}(x - 2aT)$ ✓

$$Ty - aT^3 = -x + 2aT$$
 ✓
$$x + Ty = 2aT + aT^3$$
 ✓

(b)(i) $\angle APC = \angle ABC$ (angles at circumference standing on same arc AC) ✓

(ii) Let $\angle APC = \angle ABC = \alpha$

$\angle ADE = \alpha$ (exterior $\angle$ of cyclic quad $ADEB$) ✓

$\angle APK = 180^\circ - \alpha$ (straight $\angle$) ✓

$\angle APK + \angle ADE = 180^\circ - \alpha + \alpha$

$$= 180^\circ$$

$\therefore$ ADKP is a cyclic quadrilateral since interior opposite angles are supplementary. ✓

(c)(i) $T = S + Ae^{-kt}$

$T - S = Ae^{-kt}$ ——— ①

$$\frac{dT}{dt} = \frac{d}{dt}(S + Ae^{-kt})$$

$$= -kAe^{-kt}$$
 ✓
$$= -k(T - S) \text{ from ①}$$

(ii) (a) when $t = 0$, $S = 20$, $T = 80$.

$$80 = 20 + Ae^0$$

$$A = 60$$
 ✓

when $t = 10$, $S = 20$, $T = 40$.

$$40 = 20 + 60e^{-10k}$$
 ✓

$$e^{-10k} = \frac{1}{3}$$

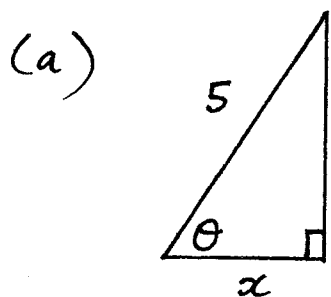
$$k = -\frac{1}{10} \ln \frac{1}{3} \quad \text{OR} \quad \frac{1}{10} \ln 3$$
 ✓

(b) when $t = 15$, $T = 20 + 60e^{-15k}$

$$\doteq 31.5^\circ$$
 ✓

(nearest tenth of a degree)

QUESTION FOUR



$$\frac{dx}{dt} = 0.02 \text{ m/s}$$

$$\frac{x}{5} = \cos \theta$$

$$\therefore \theta = \cos^{-1} \frac{x}{5}$$

$$\therefore \frac{d\theta}{dx} = \frac{-1}{\sqrt{25-x^2}}$$

By the chain rule, $\frac{d\theta}{dt} = \frac{d\theta}{dx} \cdot \frac{dx}{dt}$

$$= \frac{-0.02}{\sqrt{25-x^2}}$$

When $x=3$,

$$\frac{d\theta}{dt} = \frac{-0.02}{4}$$

$$= -\frac{1}{200} \text{ or } -0.005 \text{ rad/s}$$

(b) The general term is ${}^{10}C_r \cdot (x^2)^{10-r} \cdot \left(\frac{2}{x}\right)^r$

$$= {}^{10}C_r \cdot x^{20-2r} \cdot 2^r \cdot x^{-r}$$

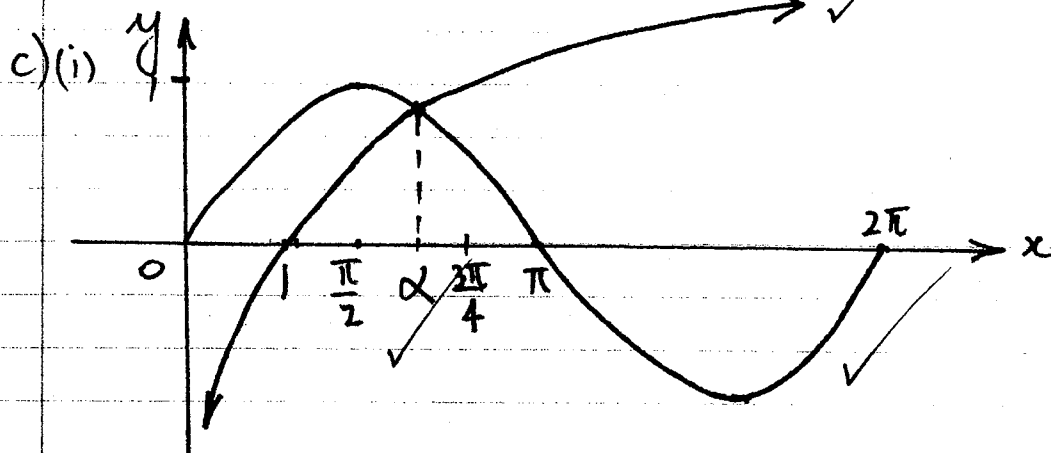
$$= {}^{10}C_r \cdot 2^r \cdot x^{20-3r}$$

We require $20-3r=2$

i.e. $r=6$

So the coefficient of x^2 is

$${}^{10}C_6 \cdot 2^6 = 13440$$



(ii) α is marked on graph.

(iii) $\ln \frac{\pi}{2} - \sin \frac{\pi}{2} \doteq -0.548 \dots$

$\ln \frac{3\pi}{4} - \sin \frac{3\pi}{4} \doteq 0.1499 \dots$

Since $f(\frac{\pi}{2}) < 0$ and $f(\frac{3\pi}{4}) > 0$, $\frac{\pi}{2} < \alpha < \frac{3\pi}{4}$

(iv) $f'(x) = \frac{1}{x} - \cos x$

So $x_2 = \frac{5\pi}{8} - \frac{f(\frac{5\pi}{8})}{f'(\frac{5\pi}{8})}$

$\doteq 2.24$

QUESTION FIVE

(a)(i) $\angle ADE = \theta + \alpha$ (exterior angle of $\triangle BCD$) ✓
 $\therefore \angle AED = \theta + \alpha$ (base angles of isosceles $\triangle ADE$) ✓
 $\therefore \angle EAB = \angle AED - \angle ABE$ (exterior angle of $\triangle ABE$) ✓
 $\quad = (\theta + \alpha) - \theta$
 $\quad = \alpha$

(ii) In triangles ABE and CBD :

$$\angle ABE = \angle CBD \text{ (given)}$$

$$\angle EAB = \angle DCB \text{ (from part (i))}$$

$$\therefore \triangle ABE \parallel \triangle CBD \text{ (A.A.)}$$

(iii) $\frac{AE}{CD} = \frac{BE}{BD}$ (matching sides are in the same ratio) ✓

$$\therefore AE \times BD = BE \times CD$$

But $AE = BD$ (given),
so $AE^2 = BE \times CD$ ✓

$$(b)(i) \quad LHS^2 = (2n+3)^2$$

$$= 4n^2 + 12n + 9$$

$$RHS^2 = 4(n+1)(n+2)$$

$$= 4n^2 + 12n + 8$$

$$\therefore LHS^2 - RHS^2 = 1 > 0$$

$$\therefore LHS^2 > RHS^2$$

$$\therefore LHS > RHS \quad (\text{since LHS and RHS are both positive for } n > 0)$$

$$(ii) \quad \text{When } n=1, \quad LHS = \frac{1}{\sqrt{1}} = 1$$

$$\text{and } RHS = 2(\sqrt{2}-1) \doteq 0.8$$

$$\therefore LHS > RHS, \text{ so the result is true for } n=1.$$

Suppose that the result is true for the positive integer $n=k$,

$$\text{i.e. suppose that } 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{k}} > 2(\sqrt{k+1}-1) \quad (*)$$

Prove that the result is true for $n=k+1$,

$$\text{i.e. prove that } 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} > 2(\sqrt{k+2}-1).$$

$$LHS = 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}}$$

$$> 2(\sqrt{k+1}-1) + \frac{1}{\sqrt{k+1}} \quad (\text{using } *)$$

$$= \frac{2[(k+1)-\sqrt{k+1}] + 1}{\sqrt{k+1}}$$

$$= \frac{2k+3-2\sqrt{k+1}}{\sqrt{k+1}} \quad \checkmark$$

$$> \frac{2\sqrt{(k+1)(k+2)} - 2\sqrt{k+1}}{\sqrt{k+1}} \quad (\text{using part (i)}) \quad \checkmark$$

$$= 2(\sqrt{k+2}-1)$$

= RHS, so the result is true for $n=k+1$ if it is true for $n=k$.

But the result is true for $n=1$, so, by induction, it is true for all

QUESTION SIX

a) $x = \tan \theta$
 $\frac{dx}{d\theta} = \sec^2 \theta$
 $dx = \sec^2 \theta d\theta$

When $x = 0$, $\tan \theta = 0$
 $\therefore \theta = 0$
 $x = 1$, $\tan \theta = 1$
 $\therefore \theta = \frac{\pi}{4}$

$$\begin{aligned} \int_0^1 \frac{1}{(1+x^2)^{3/2}} dx &= \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)^{3/2}} \quad \checkmark \checkmark \\ &= \int_0^{\pi/4} \frac{d\theta}{\sec \theta} \quad \checkmark \\ &= [\sin \theta]_0^{\pi/4} \\ &= \sin \frac{\pi}{4} - 0 \\ &= \frac{1}{\sqrt{2}} \quad \checkmark \end{aligned}$$

b) $(3+2x)^n$ has general term ${}^nC_r 3^{n-r} 2^r x^r$. $\checkmark$

When $r = 5$ and $r = 6$ we have:

$${}^nC_5 3^{n-5} 2^5 = {}^nC_6 3^{n-6} 2^6 \quad \checkmark$$

$$\frac{n!}{5!(n-5)!} \cdot 3 = \frac{n!}{6!(n-6)!} \cdot 2 \quad \checkmark$$

$$6 \times 3 = 2 \times (n-5)$$

$$18 = 2n - 10$$

$$\therefore n = 14. \quad \checkmark$$

$$c) \left. \begin{aligned} x_o &= Ut \cos \alpha \\ y_o &= Ut \sin \alpha - \frac{1}{2}gt^2 \end{aligned} \right\} \begin{aligned} x_A &= Vt \cos \beta \\ y_A &= h + Vt \sin \beta - \frac{1}{2}gt^2 \end{aligned}$$

Particles collide when $x_o = x_A$:

$$\begin{aligned} Ut \cos \alpha &= Vt \cos \beta \\ U \cos \alpha &= V \cos \beta \quad \text{--- ①} \quad \checkmark \end{aligned}$$

When $y_o = y_A$ and $t = T$:

$$UT \sin \alpha - \frac{1}{2}gT^2 = h + VT \sin \beta - \frac{1}{2}gT^2 \quad \text{--- ②} \quad \checkmark$$

$$T(U \sin \alpha - V \sin \beta) = h$$

$$T = \frac{h}{U \sin \alpha - V \sin \beta}$$

$$= \frac{h}{U \sin \alpha - \frac{U \cos \alpha}{\cos \beta} \cdot \sin \beta} \quad \checkmark \quad \text{from ①}$$

$$= \frac{h}{\frac{U \sin \alpha \cos \beta - U \cos \alpha \sin \beta}{\cos \beta}}$$

$$= \frac{h \cos \beta}{U (\sin \alpha \cos \beta - \cos \alpha \sin \beta)}$$

$$= \frac{h \cos \beta}{U \sin (\alpha - \beta)} \quad \checkmark$$

$$\begin{aligned}
 7. (a) (i) & (1+x)^{10n} + (1-x)^{10n} \\
 &= 1 + \binom{10n}{1}x + \binom{10n}{2}x^2 + \dots + \binom{10n}{10n-1}x^{10n-1} + x^{10n} \\
 &\quad + 1 - \binom{10n}{1}x + \binom{10n}{2}x^2 - \dots - \binom{10n}{10n-1}x^{10n-1} + x^{10n} \\
 &= 2 \left[1 + \binom{10n}{2}x^2 + \dots + \binom{10n}{10n-2}x^{10n-2} + x^{10n} \right]
 \end{aligned}$$

(ii) Let $x=1$ and $n=3$

$$\text{So } (1+1)^{30} + 0^{30} = 2 \left[1 + \binom{30}{2} + \binom{30}{4} + \dots + \binom{30}{28} + \binom{30}{30} \right]$$

$$\text{i.e. } 1 + \binom{30}{2} + \binom{30}{4} + \dots + \binom{30}{28} + \binom{30}{30} = 2^{29}$$

$$(b) (i) f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\begin{aligned}
 \Rightarrow f(-x) &= \frac{e^{-x} - e^x}{e^{-x} + x} \\
 &= - \frac{e^x + e^{-x}}{e^x + e^{-x}}
 \end{aligned}$$

$$\Rightarrow f(-x) = -f(x)$$

Hence $f(x)$ is an odd function.

$$(ii) f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\text{So } f(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$

$$\text{As } x \rightarrow \infty, e^{-2x} \rightarrow 0$$

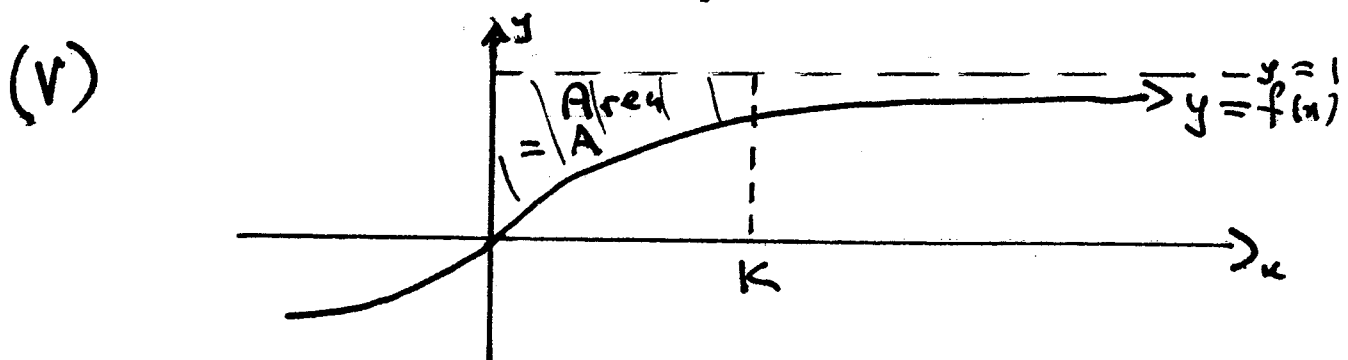
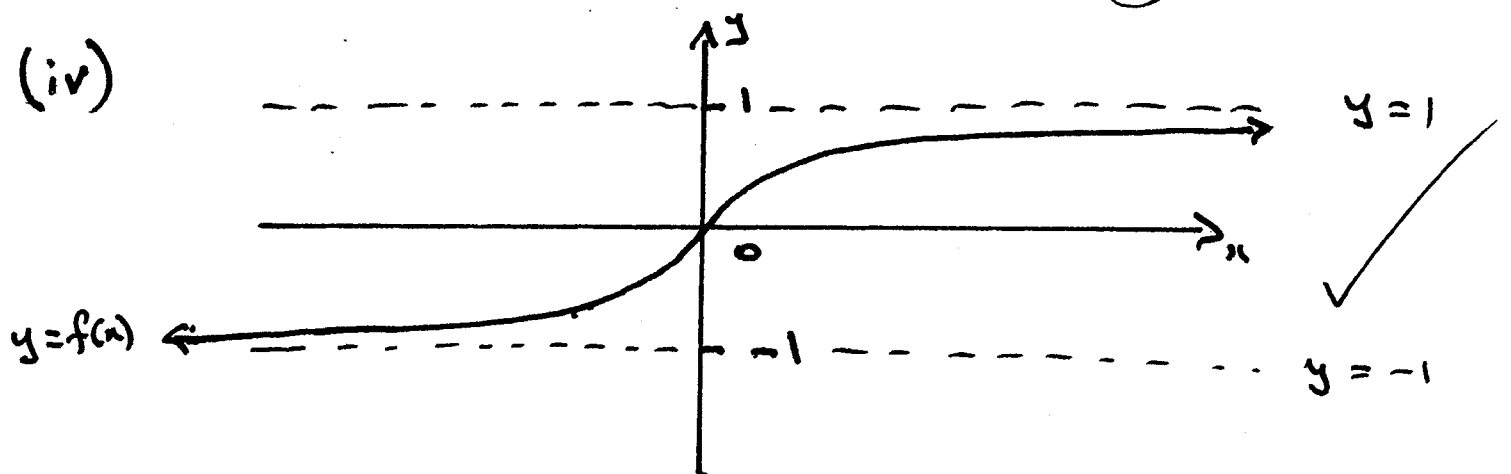
$$\text{So as } x \rightarrow \infty, f(x) \rightarrow 1$$

Furthermore as $f(x)$ is odd then

$$\text{as } x \rightarrow -\infty, f(x) \rightarrow -1$$

$$\begin{aligned}
 \text{(iii)} \quad f(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} \\
 &= \frac{e^{2x} - 1}{e^{2x} + 1} \\
 \Rightarrow f'(x) &= \frac{2e^{2x}(e^{2x} + 1) - 2e^{2x}(e^{2x} - 1)}{(e^{2x} + 1)^2} \\
 &= \frac{4e^{2x}}{(e^{2x} + 1)^2} \quad \checkmark
 \end{aligned}$$

But $e^{2x} > 0$ for all x .
 So $f'(x) > 0$ for all x . } ✓



$$\begin{aligned}
 A &= k - \int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx \\
 &= k - \left[\ln(e^x + e^{-x}) \right]_0^k \quad \checkmark \\
 &= k - \ln\left(e^k + \frac{1}{e^k}\right) + \ln 2
 \end{aligned}$$

$$\begin{aligned}
 \text{So } A &= k - \ln\left(\frac{e^{2k} + 1}{e^k}\right) + \ln 2 \\
 &= k - \ln(e^{2k} + 1) + \ln e^k + \ln 2 \\
 &= 2k - \ln(e^{2k} + 1) + \ln 2.
 \end{aligned}$$

Now as both $\ln x$ and e^{2x} are increasing functions then $\ln(e^{2k} + 1) > \ln e^{2k} = 2k$

Hence $2k - \ln(e^{2k} + 1) < 0$ for all $k > 0$

Hence $A < \ln 2$ as required.